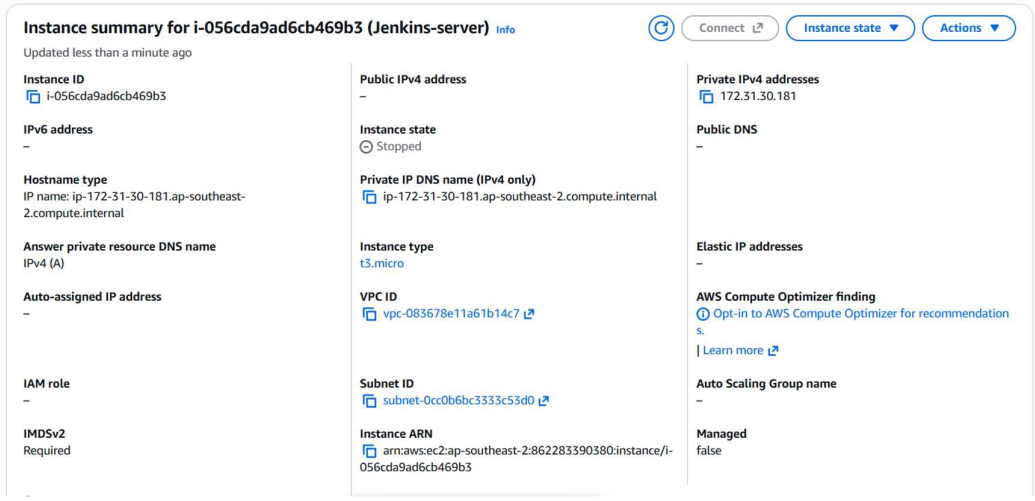
Linux Challenge

1. Launch one aws ec2 and install jenkins.

Create a AWS EC2 Instance



Instance summary for i-056cda9ad6cb469b3 (Jenkins-server) Info

Connect | Instance state | Actions

Updated less than a minute ago

Instance ID i-056cda9ad6cb469b3	**Public IPv4 address** –	**Private IPv4 addresses** 172.31.30.181
IPv6 address –	**Instance state** Stopped	**Public DNS** –
Hostname type IP name: ip-172-31-30-181.ap-southeast-2.compute.internal	**Private IP DNS name (IPv4 only)** ip-172-31-30-181.ap-southeast-2.compute.internal	
Answer private resource DNS name IPv4 (A)	**Instance type** t3.micro	**Elastic IP addresses** –
Auto-assigned IP address –	**VPC ID** vpc-083678e11a61b14c7	**AWS Compute Optimizer finding** Opt-in to AWS Compute Optimizer for recommendations. Learn more
IAM role –	**Subnet ID** subnet-0cc0b6bc3333c53d0	**Auto Scaling Group name** –
IMDSv2 Required	**Instance ARN** arn:aws:ec2:ap-southeast-2:862283390380:instance/i-056cda9ad6cb469b3	**Managed** false

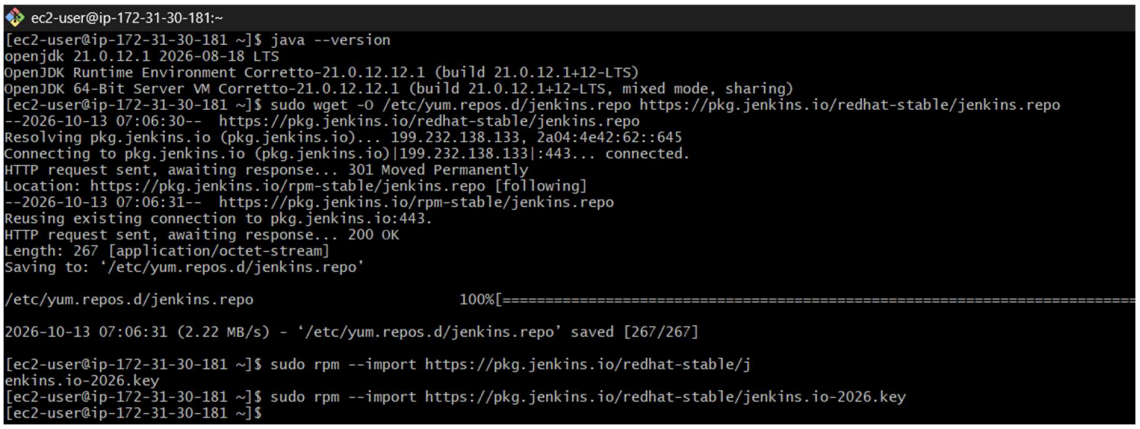
Then installed java 21 version

```
[ec2-user@ip-172-31-30-181 ~]$ sudo dnf update -y
Last metadata expiration check: 0:11:41 ago on Tue Oct 13 06:41:48 2026.
Dependencies resolved.
Nothing to do.
Complete!
[ec2-user@ip-172-31-30-181 ~]$ sudo dnf install java-21-amazon-corretto -y
Last metadata expiration check: 0:12:02 ago on Tue Oct 13 06:41:48 2026.
Dependencies resolved.
================================================================================
 Package                         Arch   Version                      Repository   Size
================================================================================
Installing:
 java-21-amazon-corretto         x86_64 1:21.0.12+12-1.amzn2023.1    amazonlinux 218 k
Installing dependencies:
 alsa-lib                        x86_64 1.2.7.2-1.amzn2023.0.4       amazonlinux 503 k
 cairo                           x86_64 1.18.0-4.amzn2023.0.3        amazonlinux 717 k
 dejavu-sans-fonts               noarch 2.37-16.amzn2023.0.2         amazonlinux 1.3 M
 dejavu-sans-mono-fonts          noarch 2.37-16.amzn2023.0.2         amazonlinux 467 k
 dejavu-serif-fonts              noarch 2.37-16.amzn2023.0.2         amazonlinux 1.0 M
 fontconfig                      x86_64 2.13.94-2.amzn2023.0.2       amazonlinux 273 k
 fonts-filesystem                noarch 1:2.0.5-12.amzn2023.0.2      amazonlinux 9.5 k
 freetype                        x86_64 2.13.2-5.amzn2023.0.2        amazonlinux 422 k
 giflib                          x86_64 5.2.1-9.amzn2023.0.4         amazonlinux  49 k
 google-noto-fonts-common        noarch 20240401-1.amzn2023.0.2      amazonlinux  17 k
 google-noto-sans-vf-fonts       noarch 20240401-1.amzn2023.0.2      amazonlinux 593 k
 graphite2                       x86_64 1.3.14-7.amzn2023.0.3        amazonlinux  97 k
 harfbuzz                        x86_64 7.0.0-2.amzn2023.0.2         amazonlinux 873 k
 java-21-amazon-corretto-headless x86_64 1:21.0.12+12-1.amzn2023.1   amazonlinux  96 M
 javapackages-filesystem         noarch 6.0.0-7.amzn2023.0.6         amazonlinux  12 k
 langpacks-core-font-en          noarch 3.0-21.amzn2023.0.4          amazonlinux  10 k
 libICE                          x86_64 1.1.1-3.amzn2023.0.1         amazonlinux  76 k
 libSM                           x86_64 1.2.4-3.amzn2023.0.1         amazonlinux  45 k
 libX11                          x86_64 1.8.10-2.amzn2023.0.2        amazonlinux 655 k
 libX11-common                   noarch 1.8.10-2.amzn2023.0.2        amazonlinux 144 k
 libXau                          x86_64 1.0.11-6.amzn2023.0.1        amazonlinux  33 k
 libXext                         x86_64 1.3.6-1.amzn2023.0.1         amazonlinux  42 k
 libXi                           x86_64 1.8.2-1.amzn2023.0.1         amazonlinux  42 k
 libXinerama                     x86_64 1.1.5-6.amzn2023.0.1         amazonlinux  16 k
 libXrandr                       x86_64 1.5.4-3.amzn2023.0.1         amazonlinux  29 k
 libXrender                      x86_64 0.9.11-6.amzn2023.0.2        amazonlinux  28 k
 libXt                           x86_64 1.3.0-3.amzn2023.0.1         amazonlinux 183 k
 libXtst                         x86_64 1.2.5-1.amzn2023.0.1         amazonlinux  22 k
 libbrotli                       x86_64 1.1.0-1.amzn2023.0.1         amazonlinux 334 k
 libjpeg-turbo                   x86_64 2.1.4-2.amzn2023.0.5         amazonlinux 190 k
 libpng                          x86_64 2:1.6.37-10.amzn2023.0.13    amazonlinux 128 k
 libxcb                          x86_64 1.17.0-1.amzn2023.0.1        amazonlinux 235 k
 pixman                          x86_64 0.43.4-1.amzn2023.0.4        amazonlinux 296 k
 xml-common                      noarch 0.6.3-56.amzn2023.0.2        amazonlinux  32 k

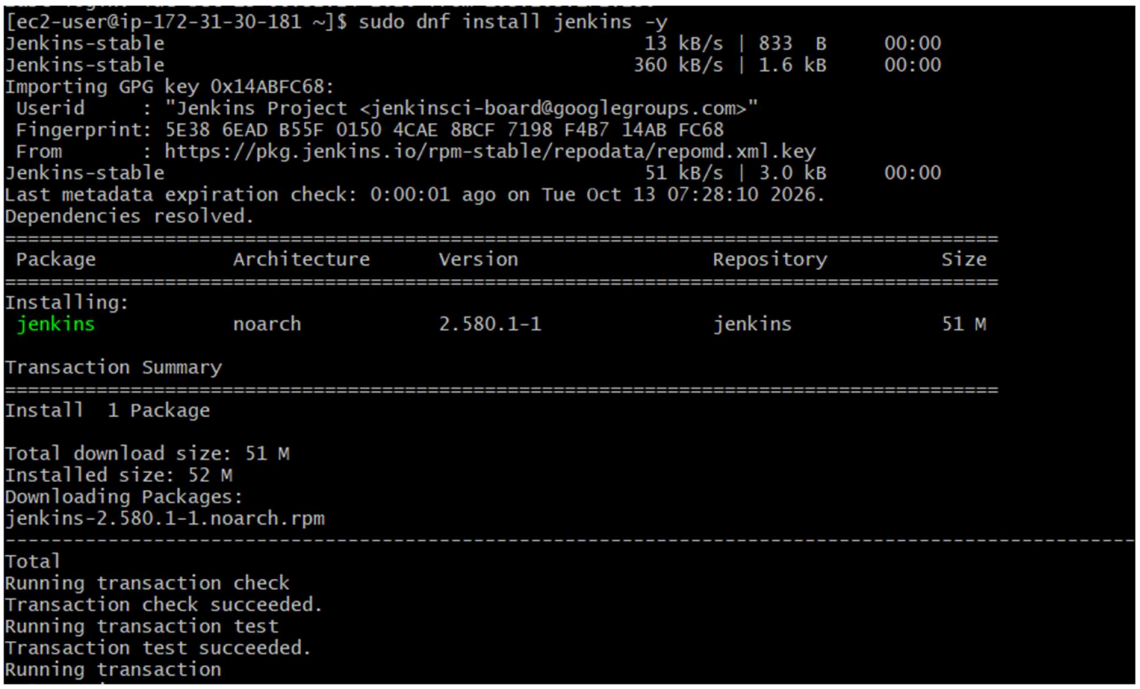
Transaction Summary
================================================================================
Install  35 Packages

Total download size: 105 M
Installed size: 283 M
Downloading Packages:
```

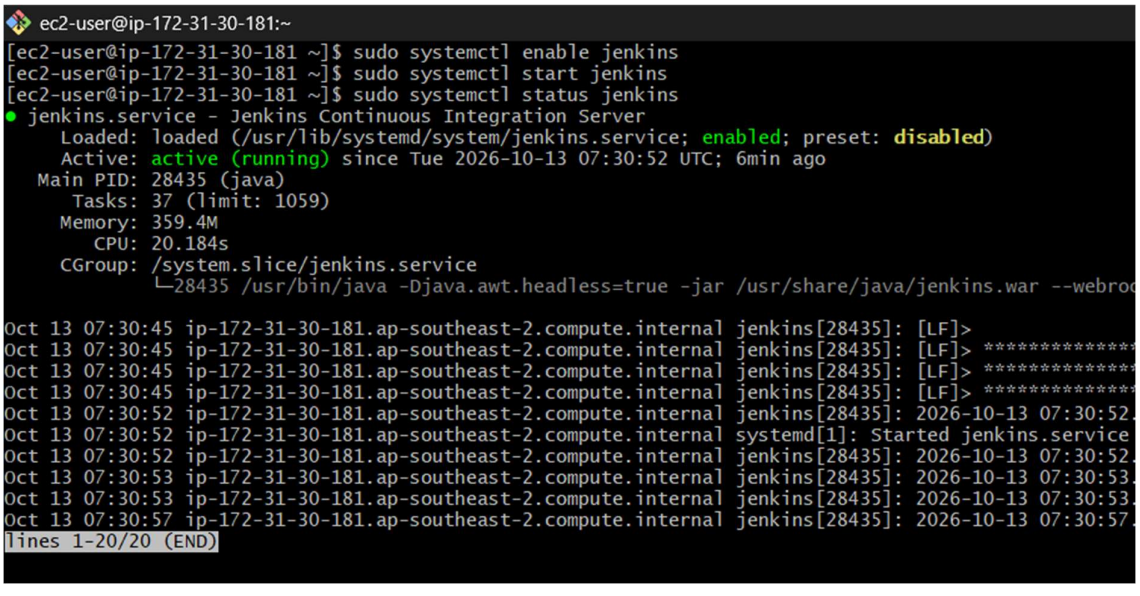
Added jenkis repo and key into the ec2 server



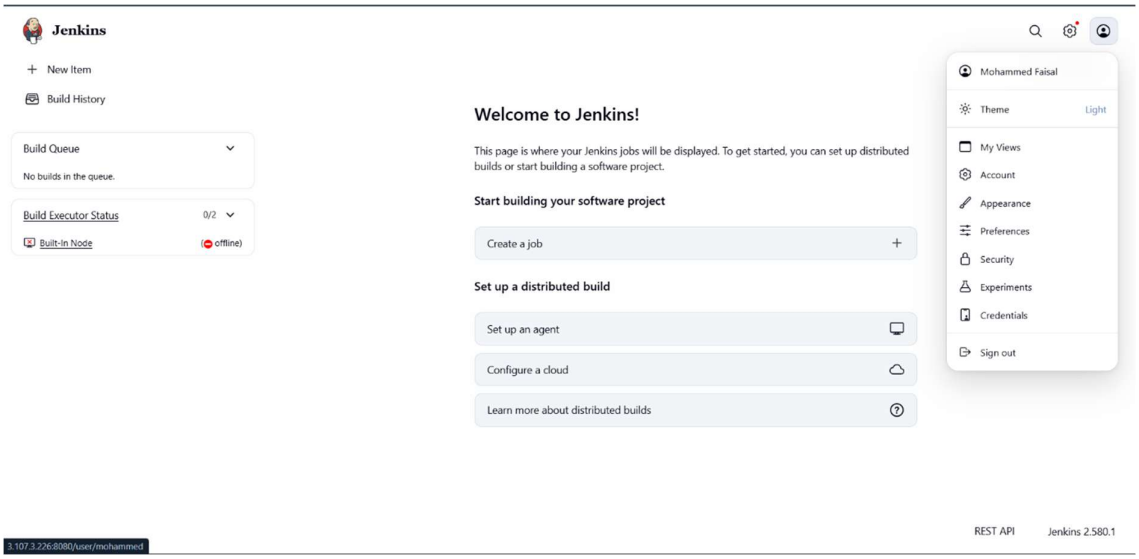
Installing Jenkins



Running the Jenkins package



Sign in into Jenkins server



2. Setup Apache Tomcat on Ubuntu Virtual Machine

Install tomcat 10 package

```
faisal@Linux:/$ sudo apt install tomcat10 -y
Installing:
  tomcat10

Installing dependencies:
  ca-certificates-java  libbcel-java             libcommons-lang3-java                libeclipse-osgi-java     openjdk-25-jre-headless
  default-jre-headless  libcommons-codec-java     libcommons-parent-java               libosgi-annotation-java  tomcat-jakartaee-migration
  java-common           libcommons-compress-java libeclipse-jdt-core-compiler-batch-java libtcnative-2           tomcat10-common
  libapache-pom-java    libcommons-io-java       libeclipse-jdt-core-java             libtomcat10-java

Suggested packages:
  default-jre  libxz-java                fonts-dejavu-extra    fonts-ipafont-mincho  | fonts-wqy-zenhei  tomcat10-admin  tomcat10-examples
  libasm-java  libosgi-annotation-java-doc  fonts-ipafont-gothic  fonts-wqy-microhei    fonts-indic        tomcat10-docs   tomcat10-user

Summary:
  Upgrading: 0, Installing: 20, Removing: 0, Not Upgrading: 433
  Download size: 77.6 MB
  Space needed: 267 MB / 16.9 GB available
```

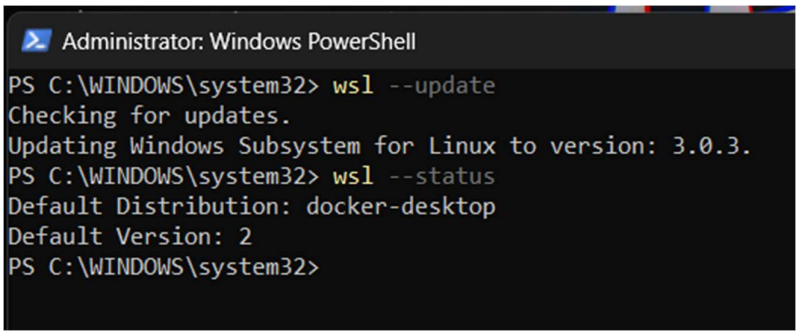
Start the tomcat server with systemctl

```
faisal@Linux:/$ sudo systemctl start tomcat10
faisal@Linux:/$ sudo systemctl status tomcat10
● tomcat10.service - Apache Tomcat 10 Web Application Server
     Loaded: loaded (/usr/lib/systemd/system/tomcat10.service; enabled; preset: enabled)
     Active: active (running) since Sat 2026-10-10 10:50:21 UTC; 3min 38s ago
 Invocation: 77fe4e0cf1504133aa884a24d44ab784
       Docs: https://tomcat.apache.org/tomcat-10.0-doc/index.html
    Process: 12815 ExecStartPre=/usr/libexec/tomcat10/tomcat-update-policy.sh (code=exited, status=0/SUCCESS)
   Main PID: 12820 (java)
      Tasks: 41 (limit: 2350)
     Memory: 149.6M (peak: 151M)
        CPU: 29.401s
     CGroup: /system.slice/tomcat10.service
             └─12820 /usr/lib/jvm/default-java/bin/java -Djava.util.logging.config.file=/var/lib/tomcat10/conf/logging.properties -Djava.util.logging>

Oct 10 10:50:25 Linux tomcat10[12820]: OpenSSL successfully initialized [OpenSSL 3.5.5 27 Jan 2026]
Oct 10 10:50:27 Linux tomcat10[12820]: Initializing ProtocolHandler ["http-nio-8080"]
Oct 10 10:50:28 Linux tomcat10[12820]: Server initialization in [3720] milliseconds
Oct 10 10:50:30 Linux tomcat10[12820]: Starting service [Catalina]
Oct 10 10:50:30 Linux tomcat10[12820]: Starting Servlet engine: [Apache Tomcat/10.1.55 (Ubuntu)]
Oct 10 10:50:30 Linux tomcat10[12820]: Deploying web application directory [/var/lib/tomcat10/webapps/ROOT]
Oct 10 10:50:36 Linux tomcat10[12820]: At least one JAR was scanned for TLDs yet contained no TLDs. Enable debug logging for this logger for a comple>
Oct 10 10:50:36 Linux tomcat10[12820]: Deployment of web application directory [/var/lib/tomcat10/webapps/ROOT] has finished in [6,007] ms
Oct 10 10:50:36 Linux tomcat10[12820]: Starting ProtocolHandler ["http-nio-8080"]
Oct 10 10:50:36 Linux tomcat10[12820]: Server startup in [7994] milliseconds
lines 1-23/23 (END)
```

3. Setup Docker Desktop on your local windows.

Using powershell and wsl to install Ubuntu



Install docker desktop and log into our account

